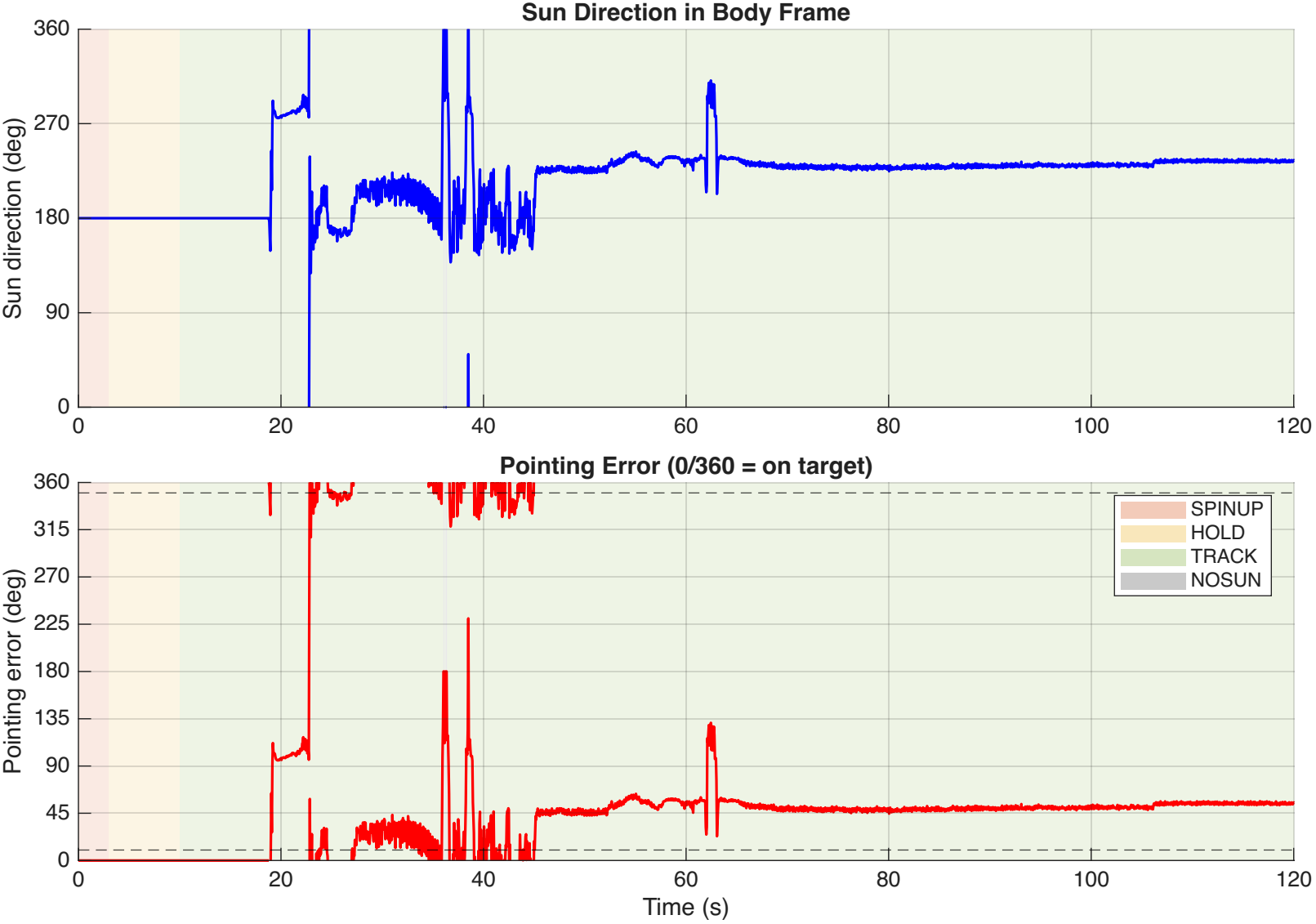
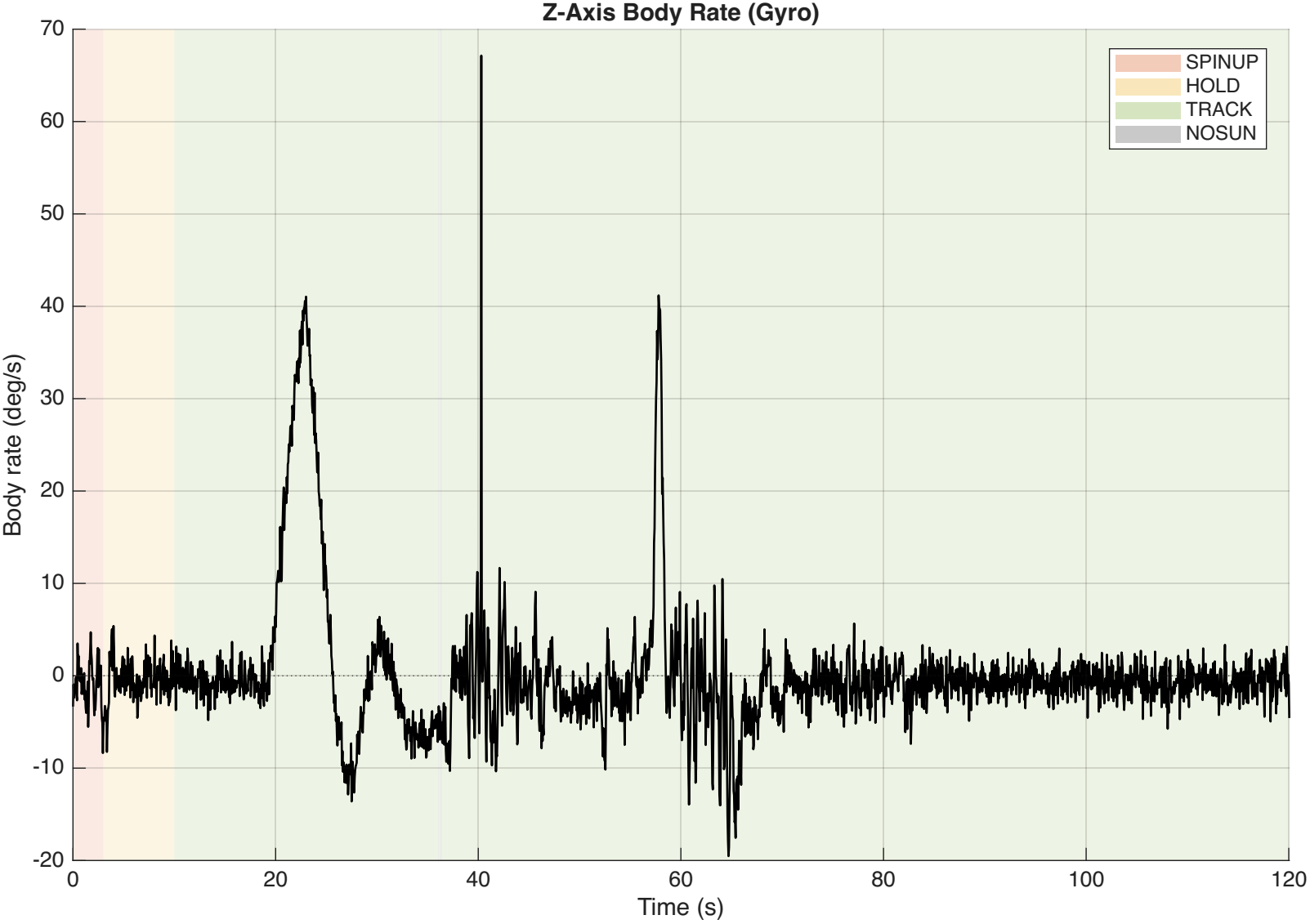
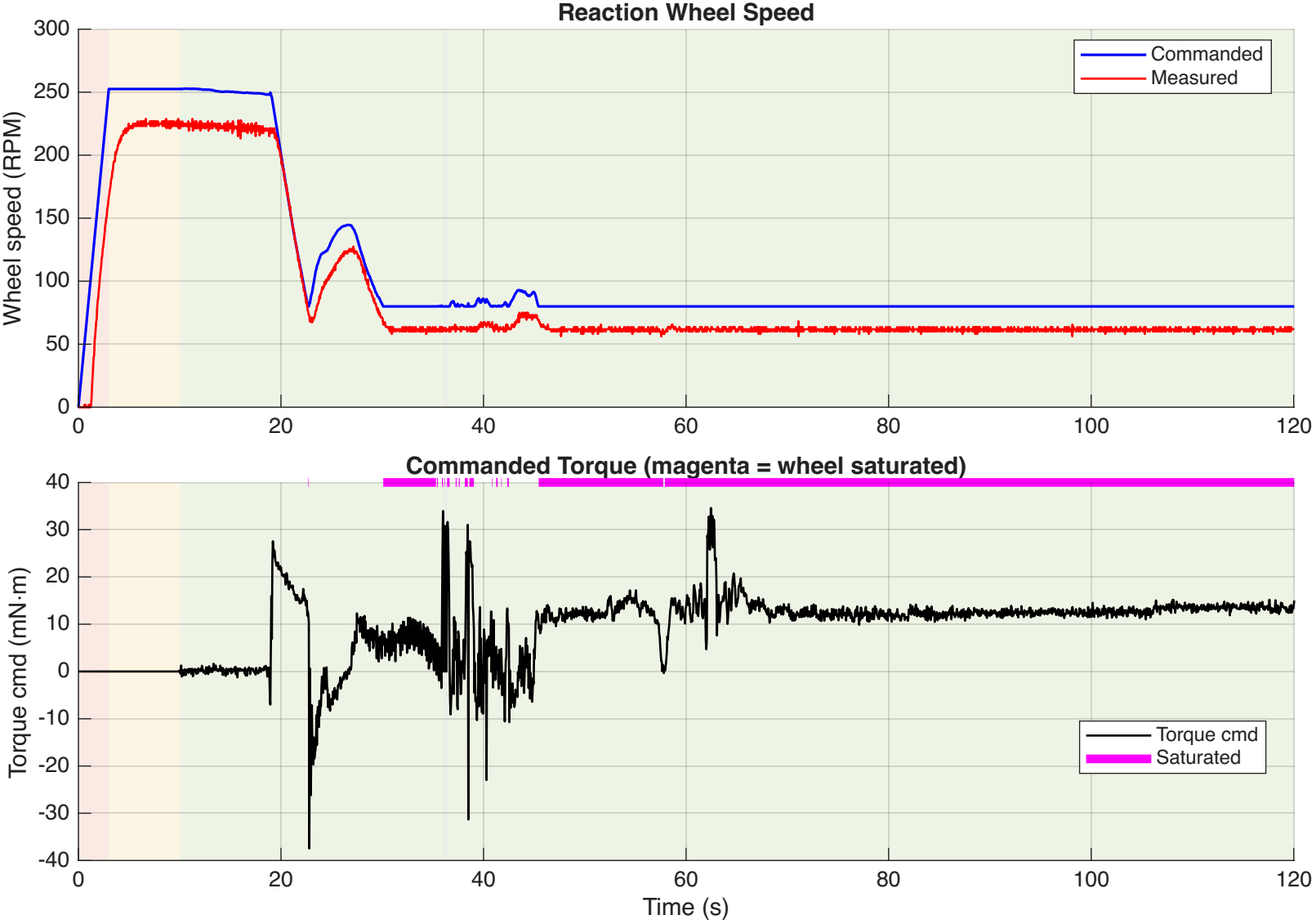
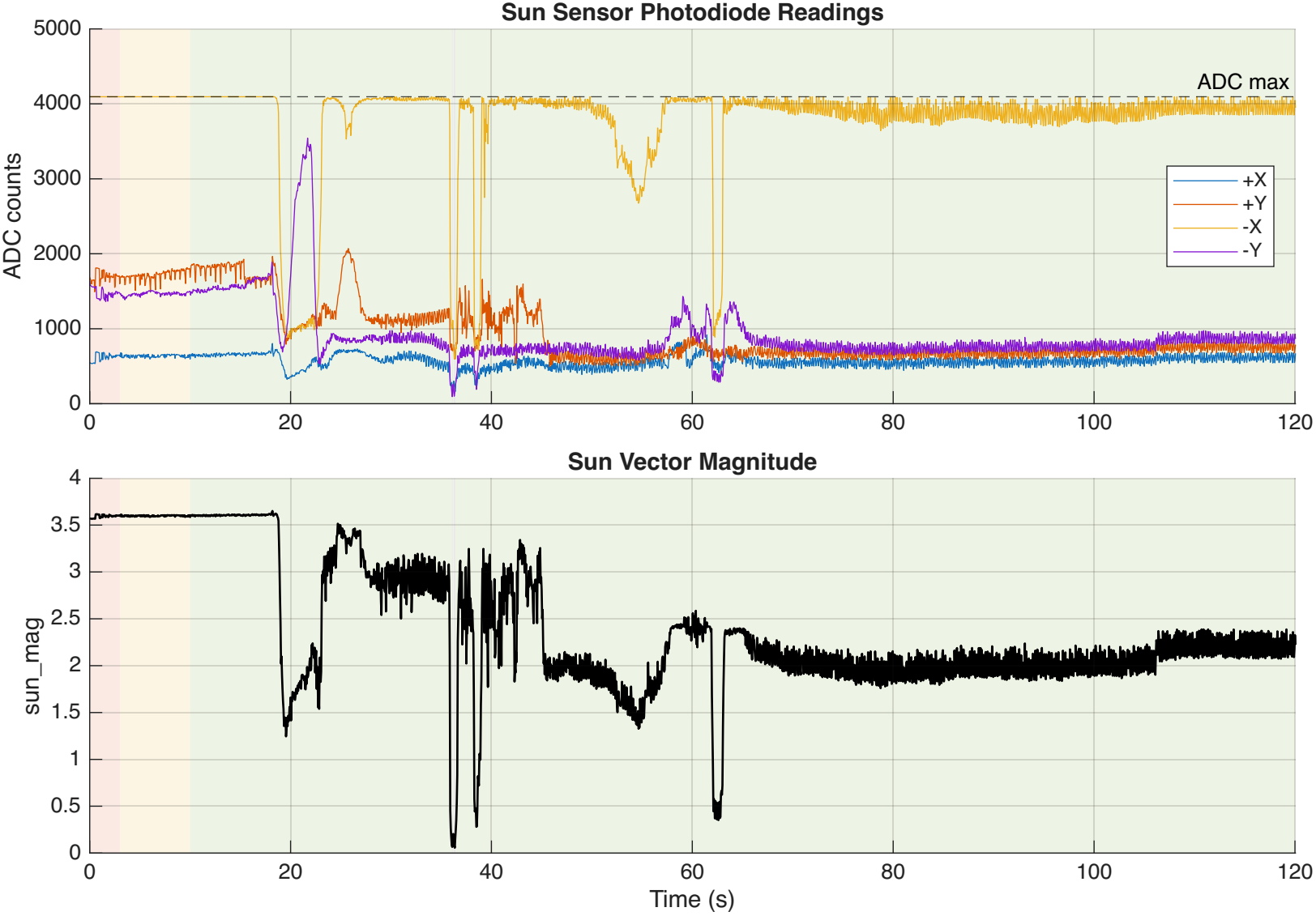
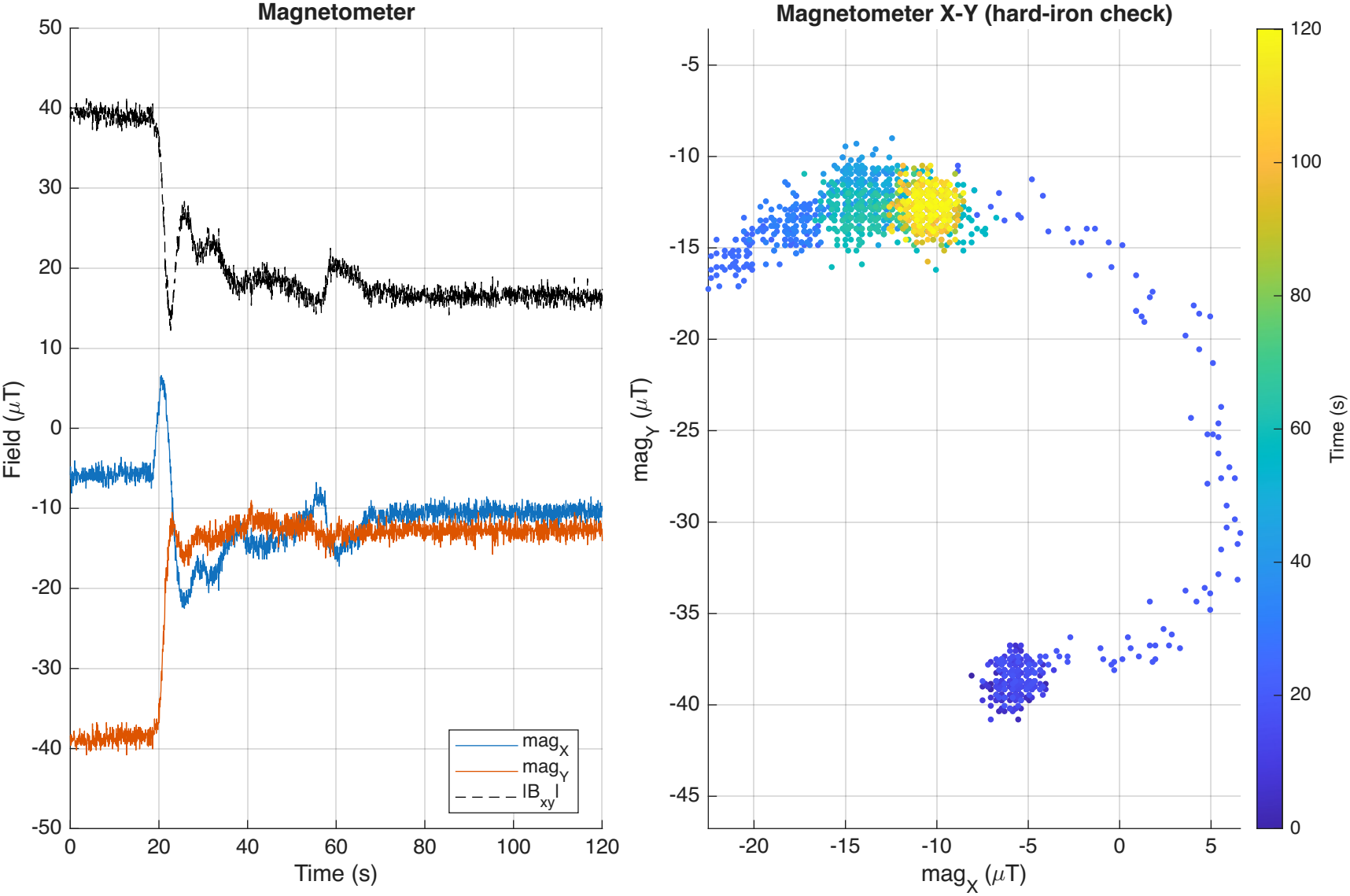
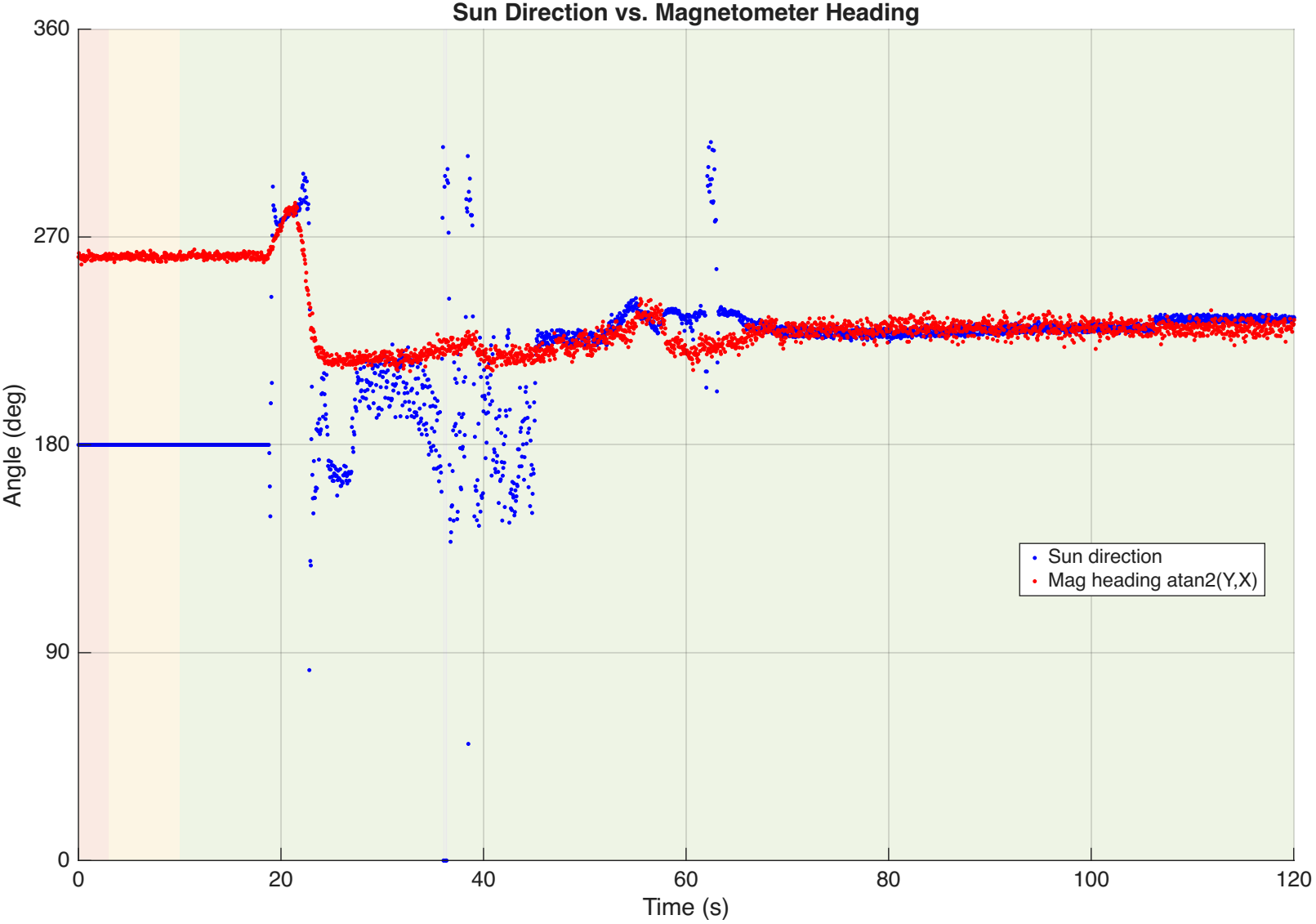

Sun Direction in Body Frame
Sun direction (deg)
360
270
180
90
0
0
20
40
60
80
100
120
Pointing Error (0/360 = on target)
Pointing error (deg)
360
315
270
225
180
135
90
45
0
SPINUP
HOLD
TRACK
NOSUN
Time (s)

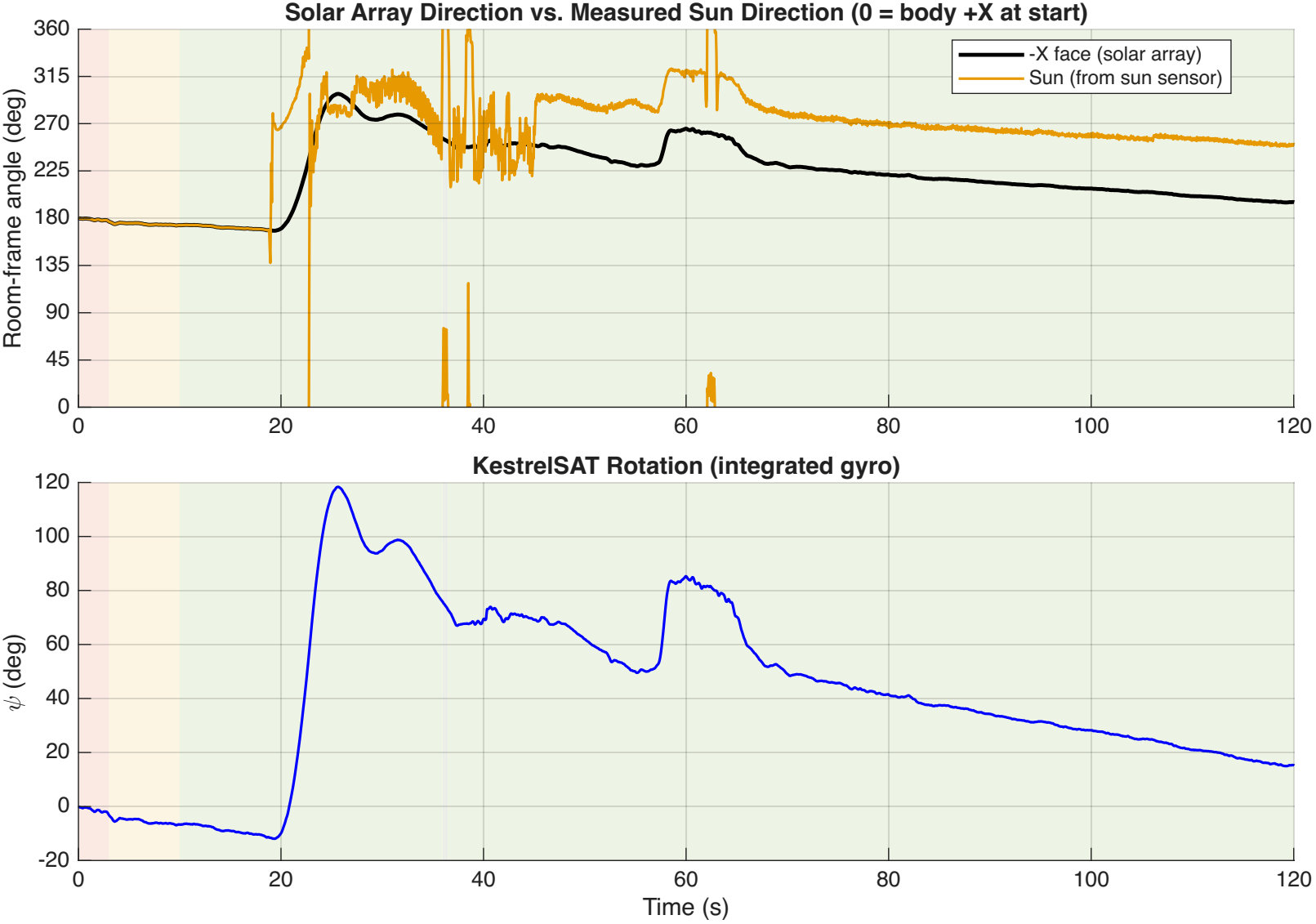

Z-Axis Body Rate (Gyro)
SPINUP
HOLD
TRACK
NOSUN
Body rate (deg/s)
Time (s)
70
60
50
40
30
20
10
0
-10
-20
0
20
40
60
80
100
120


Reaction Wheel Speed
Wheel speed (RPM)
Commanded
Measured
Commanded Torque (magenta = wheel saturated)
Torque cmd (mN·m)
Torque cmd
Saturated
Time (s)


Sun Sensor Photodiode Readings
ADC counts
ADC max
+X
+Y
-X
-Y
Sun Vector Magnitude
sun_mag
Time (s)


Magnetometer
Field (μT)
Time (s)
mag_X
mag_Y
|B_xy|
Magnetometer X-Y (hard-iron check)
mag_Y (μT)
mag_X (μT)
Time (s)


Sun Direction vs. Magnetometer Heading
360
270
180
90
0
Angle (deg)
0
20
40
60
80
100
120
Time (s)
Sun direction
Mag heading atan2(Y,X)


Solar Array Direction vs. Measured Sun Direction (0 = body +X at start)
-X face (solar array)
Sun (from sun sensor)
Room-frame angle (deg)
360
315
270
225
180
135
90
45
0
0
20
40
60
80
100
120
KestrelSAT Rotation (integrated gyro)
ψ (deg)
120
100
80
60
40
20
0
-20
0
20
40
60
80
100
120
Time (s)